

MULTIVARIATE LINEAR DISCRIMINATION WHEN THE COVARIANCE MATRIX IS UNKNOWN

Franz Streit
University of Geneva

Key words: Plug-in-rule, identification, unconditional probabilities of misclassification.

SUMMARY

Suppose that either $\underline{X} \sim \underline{X}_1$ or $\underline{X} \sim \underline{X}_2$ and that

$$\underline{X}_i \sim N_k(\underline{\mu}_i, \Sigma) \quad [i=1, 2; \underline{\mu}_1 \neq \underline{\mu}_2; |\Sigma| \neq 0]$$

where $\underline{\mu}_1$ and $\underline{\mu}_2$ are known and Σ is unknown. For the corresponding problem of identification expressions for the unconditional probabilities of misclassification $e(1:2)$ and $e(2:1)$ of a random vector $\underline{X}$ to the population of $\underline{X}_1$ resp. of $\underline{X}_2$ are derived. These error probabilities are determined under the assumptions of equal costs of misclassification, of equal a-priori probabilities, and of the application of the plug-in-rule, obtained from the linear discriminant procedure by replacing Σ by its usual estimator based on training samples, as identification rule.

1. Introduction

In many applications of discriminant analysis it is reasonable to assume that the problem consists in identifying a sample unit to one of two given normally distributed populations with different mean vectors but identical covariance matrix. This leads to the application of Fisher's linear discriminant function with its well-known optimality properties. Very often some of the parameters are unknown, but may be estimated from training samples, a procedure which amounts to replace the optimal identification rule by the corresponding plug-in-rule. Such rules have been extensively studied in the literature (see e.g. Anderson (1966), pp. 137 ff., Das Gupta (1973), pp. 92 ff. and references quoted in this paper, Kshirsagar (1972), pp. 195 ff., Lachenbruch (1975), pp. 9 ff.) but mainly for the situation where all parameters are unknown; notable exceptions are in particular John (1961) and Bandyopadhyay (1978). The case of partial information on the parameters most likely to arise in practical applications is the case where the mean vectors $\underline{\mu}_1$ and $\underline{\mu}_2$ are known but where the covariance matrix Σ is unknown and it is for this set-up that the properties of the relevant identification rule are analysed in this paper.

Decision-theoretic motivation for the use of this discrimination procedure was given in Streit (1977) and some of the results were announced in an abstract (Streit (1979)).

Let thus G_1 and G_2 be two populations with corresponding k -dimensional normally distributed random vectors $\underline{X}_1$ and $\underline{X}_2$, i.e. $\underline{X}_1 \sim N_k(\underline{\mu}_1, \Sigma)$ and $\underline{X}_2 \sim N_k(\underline{\mu}_2, \Sigma)$, where $\underline{\mu}_1$ and $\underline{\mu}_2$ are supposed to be known and different and Σ is supposed to be unknown and positive definite. A random sample

$$\underline{X}^{(1)}, \dots, \underline{X}^{(n_1)}$$

from G_1 and a random sample

$$\underline{X}^{(n_1+1)}, \dots, \underline{X}^{(n_1+n_2)}$$

from G_2 are given and it is assumed that $n := n_1 + n_2 \geq k$. Consider a random vector $\underline{X}$ about which is known that $\underline{X} \sim \underline{X}_1$ or $\underline{X} \sim \underline{X}_2$ and that the corresponding sample unit comes either from G_1 or from G_2 . An identification rule ϕ is any function defined on $R^{(n_1+n_2+1)k}$ taking its values in $[0, 1]$. Using the abbreviation $\underline{D} := (\underline{X}; \underline{X}^{(1)}, \dots, \underline{X}^{(n)})$ for the observable quantities, the values $\phi(\underline{d})$ are interpreted as the conditional probabilities of assigning the sample unit to G_1 given $\underline{D} = \underline{d}$, i.e. given the observation $\underline{X} = \underline{x}$ from the unclassified sample unit and the measurements $\underline{X}^{(i)} = \underline{x}^{(i)}$ [$i = 1, \dots, n$] obtained from a realization of the training samples. In this article we study the properties of the plug-in-rule ϕ^* for this problem, which is based on the conventional estimator

$$S := n^{-1} \left\{ \sum_{i=1}^{n_1} (\underline{X}^{(i)} - \underline{\mu}_1)(\underline{X}^{(i)} - \underline{\mu}_1)' + \sum_{i=n_1+1}^n (\underline{X}^{(i)} - \underline{\mu}_2)(\underline{X}^{(i)} - \underline{\mu}_2)' \right\}.$$

for Σ and which is obtained from the optimal Bayes rule with equal a-priori probabilities, equal costs of misclassification and all parameters known (Anderson (1966), p. 134). This rule is specified by

$$\phi^*(\underline{d}) = 1 \text{ iff } \{ \underline{x} \in R^k: (\underline{\mu}_2 - \underline{\mu}_1)' S^{-1} (\underline{x} - \frac{1}{2}(\underline{\mu}_1 + \underline{\mu}_2)) \leq 0 \}$$

and

$$\phi^*(\underline{d}) = 0 \text{ iff } \{ \underline{x} \in R^k: (\underline{\mu}_2 - \underline{\mu}_1)' S^{-1} (\underline{x} - \frac{1}{2}(\underline{\mu}_1 + \underline{\mu}_2)) > 0 \}$$

The main statistical problem in this context consists in assessing the performance of the identification rule ϕ^* .

The following symbols and notations will be used in the sequel: D_j , designating the random variable

$$D_j := (\underline{\mu}_2 - \underline{\mu}_1)' S^{-1} (X_j - \frac{1}{2} (\underline{\mu}_1 + \underline{\mu}_2)), \quad [j = 1, 2];$$

I_k , designating the $k \times k$ identity matrix; $\Delta := (\underline{\mu}_2 - \underline{\mu}_1)' \Sigma^{-1} (\underline{\mu}_2 - \underline{\mu}_1)$, designating the Mahalanobis distance between the populations G_1 and G_2 ; $W_p(\nu, \Sigma^*)$, designating the p -dimensional central Wishart distribution with ν degrees of freedom and associated matrix Σ^* ; χ_ν^2 , designating the chi-square distribution of ν degrees of freedom; $H_1 \underset{\lambda}{\wedge} H_2$, designating the mixed (compound) distribution obtained according to the formula $H(\underline{x}) = \int_{\Lambda} H_1(\underline{x}; \lambda) dH_2(\lambda)$, where $\{H_1(\cdot; \lambda); \lambda \in \Lambda\}$ is a family of distribution functions indexed by the possibly multivariate parameter λ , H_2 is a distribution function defined over Λ and H is the distribution function of the mixture $H_1 \underset{\lambda}{\wedge} H_2$; and Φ , designating the distribution function of $N_1(0, 1)$.

2. Results

The unconditional probabilities of misclassification of the sample unit to G_1 resp. to G_2 are given by

$$e(1|2) = \Pr(\phi^*(D) = 1 \mid \underline{X} = \underline{X}_2) = \Pr(D_2 \leq 0)$$

resp.

$$e(2:1) = \Pr(\phi^*(D) = 0 \mid \underline{X} = \underline{X}_1) = \Pr(D_1 > 0).$$

Noting that $n \cdot S$ follows a Wishart distribution we remark that the following invariance property is of importance for the determination of the required error probabilities.

Lemma 1

Suppose that $n \cdot S \sim W_k(n, \Sigma)$ and that independently of this

$$\underline{X}_j \sim N_k(\underline{\mu}_j, \Sigma) \quad [j = 1, 2], \text{ where } \underline{\mu}_j \in \mathbb{R}^k \text{ and } \Sigma > 0.$$

Then it follows, that the distribution of D_j depends on the parameters only in terms of Δ . More exactly we have

$$D_1 \sim D_1^* := \underline{c}_k'(\Delta) \xi^{-1} \left(\underline{z} - \frac{1}{2} \underline{c}_k(\Delta) \right)$$

and

$$D_2 \sim D_2^* := \underline{c}_k'(\Delta) \underline{\zeta}^{-1} \left(\underline{Z} + \frac{1}{2} \underline{c}_k(\Delta) \right),$$

where $\underline{c}_k(\Delta) \in R^k$ and $\underline{c}_k'(\Delta) = (0, \dots, 0, \sqrt{\Delta})$, $\underline{Z} \sim N_k(0, I_k)$, and independently of this $n\zeta \sim W_k(n, I_k)$.

Proof:

Let Ω be an orthogonal $k \times k$ -matrix such that $\Omega \Sigma^{-1/2} (\underline{\mu}_2 - \underline{\mu}_1) = \underline{c}_k(\Delta)$. Consider the following transformations

$$\underline{X}_j \rightarrow \underline{X}_j^* := \Omega \Sigma^{-1/2} \left(\underline{X}_j - \frac{1}{2} (\underline{\mu}_1 + \underline{\mu}_2) \right)$$

$$\underline{\mu}_j \rightarrow \underline{\mu}_j^* := \Omega \Sigma^{-1/2} \left(\underline{\mu}_j - \frac{1}{2} (\underline{\mu}_1 + \underline{\mu}_2) \right) \quad [j = 1, 2]$$

$$\underline{X}^{(i)} \rightarrow \underline{X}^{(i)*} := \Omega \Sigma^{-1/2} \left(\underline{X}^{(i)} - \frac{1}{2} (\underline{\mu}_1 + \underline{\mu}_2) \right) \quad [i = 1, \dots, n]$$

$$nS \rightarrow n\zeta := n \Omega \Sigma^{-1/2} S \Sigma^{-1/2} \Omega'$$

Thus $\underline{X}_1^* \sim N_k(-\frac{1}{2} \underline{c}_k(\Delta), I_k)$, $\underline{X}_2^* \sim N_k(\frac{1}{2} \underline{c}_k(\Delta), I_k)$, $n\zeta \sim W_k(n, I_k)$, and therefore

$$D_j^* \sim D_j, \quad [j = 1, 2].$$

Applying this lemma to our problem yields the result:

Theorem 1

For $k \geq 2$ the following relations hold for the unconditional error probabilities of

ϕ^* :

$$e(1:2) = \Pr(D_2^* \leq 0) = \Pr\left(Z_k + \frac{\sqrt{\Delta}}{2} \leq \sum_{i=1}^{k-1} Z_i V_i\right)$$

and

$$e(2:1) = \Pr(D_1^* > 0) = \Pr\left(Z_k - \frac{\sqrt{\Delta}}{2} > \sum_{i=1}^{k-1} Z_i V_i\right)$$

where $\underline{Z} = (Z_1, \dots, Z_k)' \sim N_k(0, I_k)$ and independently of this

$$\underline{V} = (V_1, \dots, V_{k-1})' \sim N_{k-1}(0, (n\zeta_{11})^{-1}) \wedge W_{k-1}(n, I_{k-1}).$$

For $k = 1$ the probabilities of misclassification are given by

$$e(1:2) = e(2:1) = \Phi\left(-\frac{1}{2}\sqrt{\Delta}\right).$$

Proof:

We indicate the derivation of the statement concerning $e(2:1)$. The relation for $e(1:2)$ is proved in an analogous way. Let $k \geq 2$ and consider the partition of the relevant matrices along the last line and the last column, thus

$$\xi = \begin{pmatrix} \xi_{11} & \xi_{\cdot} \\ \xi_{\cdot}' & \xi_{kk} \end{pmatrix}, \quad \xi^{-1} = \begin{pmatrix} \xi^{11} & \xi_{\cdot} \\ \xi_{\cdot}' & \xi^{kk} \end{pmatrix}, \quad \text{and } Z = \begin{pmatrix} Z_{(1)} \\ Z_k \end{pmatrix}$$

$$\text{Then } D_1^* = \sqrt{\Delta} \xi_{\cdot}' Z_{(1)} + \sqrt{\Delta} \xi^{kk} (Z_k - \frac{1}{2} \sqrt{\Delta}).$$

But according to Kshirsagar (1972), p. 117

$$\xi^{kk} = \xi_{22 \cdot 1}^{-1} \text{ with } \xi_{22 \cdot 1} = \xi_{kk} - \xi_{\cdot}' \xi_{11}^{-1} \xi_{\cdot}$$

and

$$\xi_{\cdot} = -\xi^{kk} \cdot \xi_{11}^{-1} \xi_{\cdot} \text{ and thus } \xi_{\cdot}' = -\xi_{22 \cdot 1}^{-1} \xi_{\cdot}' \xi_{11}^{-1}$$

By taking into account that $E[n\xi] = I_k$, it follows (Kshirsagar (1972), p. 113) that

$$n\xi_{22 \cdot 1} \sim \chi_{n-k+1}^2 \text{ and this independently of } B^* : \xi_{11}^{-1} \xi_{\cdot} = -\xi_{22 \cdot 1}^{-1} \xi_{\cdot}.$$

Furthermore it is known (Kshirsagar (1972), p. 64), that the conditional distribution of B^* given the realization of $n\xi_{11}$ is

$$B^* | n\xi_{11} \sim N_{k-1} \left(0, \frac{1}{n} \xi_{11}^{-1} \right) \text{ while}$$

$$n\xi_{11} \sim W_{k-1}(n, I_{k-1}).$$

Therefore we have

$$\begin{aligned} D_1^* &= -\sqrt{\Delta} \xi_{22 \cdot 1}^{-1} B^{**} Z_{(1)} + \sqrt{\Delta} \xi^{kk} (Z_k - \frac{1}{2} \sqrt{\Delta}) \\ &\sim n \sqrt{\Delta} (\chi_{n-k+1}^2)^{-1} [-V' Z_{(1)} + Z_k - \frac{1}{2} \sqrt{\Delta}], \end{aligned}$$

where V has the distribution indicated in the theorem independent of χ_{n-k+1}^2 and Z .

The events $D_1^\star > 0$ and $Z_k - \frac{1}{2}\sqrt{\Delta} > \sum_{i=1}^{k-1} Z_i V_i$ are thus equivalent, which shows the validity of the statement concerning $e(2:1)$ for $k \geq 2$.

For $k = 1$ direct computation shows that the events $D_1^\star > 0$ and $Z_k - \frac{1}{2}\sqrt{\Delta} > 0$ are equivalent, which implies $e(2:1) = \Phi(-\frac{1}{2}\sqrt{\Delta})$.

Remarks

(1) As immediate consequences of this theorem we find

(a) $e(1:2)$ and $e(2:1)$ are decreasing functions of Δ .

(b) $e(1:2) = e(2:1)$.

(2) For the univariate case only, a different derivation of the unconditional probabilities of misclassification is given in the report abstracted in Hug and Streit (1976).

For $k \geq 2$ the knowledge of the density function $f_{L_{k,n}}$ of the random variable $L_{k,n} := Z_k - \tilde{Z}_{(1)}' \tilde{V}$ allows to determine $e(1:2)$ and $e(2:1)$. Lemma 2-4 are derived for this purpose using the following additional symbols and notation: $\mathbb{N}$, designating the set of all natural numbers; $F_{\nu_1, \nu_2}^\star$, designating the canonical Fisher-

Snedecor distribution, i.e. $F_{\nu_1, \nu_2}^\star := \frac{x_{\nu_1}^2}{x_{\nu_2}^2}$, the variables in the numerator and in the de-

nominator being independent; F , designating the hypergeometric function; $W_{\kappa, \mu}$, designating Whittaker's W -function; $\Gamma^\star$, designating the incomplete Gamma-function, defined by $\Gamma^\star(a, x) := \int_x^\infty e^{-t} t^{a-1} dt$ [$x \geq 0$]; Γ , designating the (complete) Gamma-function; and B , designating the Beta-function.

Lemma 2

Let $\tilde{Z}_{(1)} \sim N_{k-1}(\underline{0}, I_{k-1})$ and independently of this

$$\tilde{V} \sim N_{k-1}(\underline{0}, U^{-1}) \bigwedge_U W_{k-1}(n, I_{k-1}).$$

Then the distribution of $\tilde{Z}_{(1)}' \tilde{V}$ is a scale-mixture of an univariate normal distribution with a $F^\star$ -distribution; more exactly the following statement holds for $k \in \mathbb{N} - \{1\}$ and $n \geq k-1$:

$$\underline{Z}_{(1)}' V \sim N_1(0, \sigma^2) \wedge F_{k-1, n-k+2}^*$$

Proof:

The conditional distribution of $\underline{Z}_{(1)}' V$ given $\underline{Z}_{(1)} = \underline{z}_{(1)}$ and $U = u$ is the normal distribution with mean 0 and variance $\underline{z}_{(1)}' u^{-1} \underline{z}_{(1)}$. According to Kshirsagar (1972), p. 67 and taking into account that $\underline{Z}_{(1)}$ is distributed independently of U , we find that

$$[\underline{Z}_{(1)}' U \underline{Z}_{(1)}]^{-1} \cdot [\underline{Z}_{(1)}' \underline{z}_{(1)}] \sim \chi_{n-(k-2)}^2$$

and this independently of $\underline{Z}_{(1)}$. This implies $\underline{Z}_{(1)}' U^{-1} \underline{Z}_{(1)} \sim F_{k-1, n-(k-2)}^*$. The lemma is obtained by combination of these results.

Lemma 3

The density function $f_{H_{k_1, k_2}}$ of the distribution

$H_{k_1, k_2} := N(0, \sigma^2) \wedge F_{k_1, k_2}^*$ is for $k_1, k_2 \in \mathbb{N}$ given by the formula

$$f_{H_{k_1, k_2}}(t) = \{(2\pi)^{1/2} B[\frac{k_1}{2}, \frac{k_2}{2}]\}^{-1} \Gamma(\frac{k_2+1}{2}) (\frac{t}{2})^{\frac{k_1-3}{4}} \\ \exp[\frac{t^2}{4}] W_{-\frac{k_1}{4} - \frac{k_2}{2} + \frac{1}{4}, -\frac{k_1}{4} + \frac{1}{4}}(\frac{t^2}{2})$$

Proof:

$$f_{H_{k_1, k_2}}(t) = \{\sqrt{2\pi} B[\frac{k_1}{2}, \frac{k_2}{2}]\}^{-1} \int_0^\infty \exp[-\frac{1}{2} \frac{t^2}{\sigma^2}] (\sigma^2)^{\frac{k_1-3}{2}} (1+\sigma^2)^{-\frac{k_1+k_2}{2}} d\sigma^2 \\ = \{\sqrt{2\pi} B[\frac{k_1}{2}, \frac{k_2}{2}]\}^{-1} \int_0^\infty \exp[-\frac{1}{2} t^2 u] u^{\frac{k_2-1}{2}} (1+u)^{-\frac{k_1+k_2}{2}} du.$$

The statement follows by application of Erdélyi et al (1953), p. 255, formula (2) and p. 264, formula (4).

Lemma 4

The density function of $L_{k,n}$ for $n \geq k \geq 2$ is given by the formula

$$f_{L_{k,n}}(t) = \{2\pi B[\frac{k-1}{2}, \frac{n-k+2}{2}]\}^{-1} \Gamma(\frac{n-k+3}{2}) \\ \exp[-\frac{1}{2}t^2] \sum_{\mu=0}^{\infty} \left\{ \frac{t^{2\mu} 2^{\mu+1/2} \Gamma(\mu+\frac{3}{2}) \Gamma(\mu+\frac{k}{2}+\frac{1}{2})}{(2\mu)! \Gamma(\mu+\frac{n}{2}+2)} \right\}.$$

Proof:

Since Z_k and $Z_{(1)}' \underline{V}$ are independent random variables we find, using lemma 3 with $k_1 = k-1$ and $k_2 = n-k+2$,

$$f_{L_{k,n}}(t) = \int_{-\infty}^{+\infty} (2\pi)^{-1/2} \exp[-\frac{1}{2}(t+u)^2] f_{H_{k-1,n-k+2}}(u) du \\ = \rho_{n,k}(t) \cdot \int_{-\infty}^{+\infty} \exp[-\frac{1}{4}u^2 - tu] \left(\frac{u}{2}\right)^{\frac{k}{4}-1} W_{\frac{k}{4}-\frac{n}{2}-\frac{1}{2}, -\frac{k}{4}+\frac{1}{2}}\left(\frac{u}{2}\right) du \\ = \rho_{n,k}(t) \cdot \sum_{\mu=0}^{\infty} \frac{t^{2\mu} 2^{\mu+\frac{1}{2}}}{(2\mu)!} \int_0^{\infty} \exp[-\frac{1}{2}\tau] \tau^{\frac{k}{4}+\mu-\frac{1}{2}} W_{\frac{k}{4}-\frac{n}{2}-\frac{1}{2}, -\frac{k}{4}+\frac{1}{2}}(\tau) d\tau$$

$$\text{with } \rho_{n,k}(t) := \{2\pi B[\frac{k-1}{2}, \frac{n-k+2}{2}]\}^{-1} \Gamma(\frac{n-k+3}{2}) \exp[-\frac{1}{2}t^2].$$

From formula 7.621.3 in Gradshteyn, I.S. and Ryzhik, I.M. (1965), p. 860 follows

$$f_{L_{k,n}}(t) = \rho_{n,k}(t) \cdot \sum_{\mu=0}^{\infty} \left\{ \frac{t^{2\mu} 2^{\mu+\frac{1}{2}} \Gamma(\mu+\frac{3}{2}) \Gamma(\mu+\frac{k}{2}+\frac{1}{2})}{(2\mu)! \Gamma(\mu+\frac{n}{2}+2)} \right. \\ \left. \cdot F\left(\mu+\frac{3}{2}, \frac{n}{2}-\frac{k}{2}+\frac{3}{2}, \mu+\frac{n}{2}+2, 0\right) \right\}.$$

This proves the validity of the statement, since

$$F\left(\mu+\frac{3}{2}, \frac{n}{2}-\frac{k}{2}+\frac{3}{2}, \mu+\frac{n}{2}+2, 0\right) = 1.$$

With the aid of these auxiliary results it is possible to derive also for $k \geq 2$ an explicit expression for the probabilities of misclassification.

Theorem 2

The unconditional probabilities of misclassification of the identification rule ϕ^* are for $n \geq k \geq 2$ given by

$$e(1:2) = e(2:1) = \{(2\pi) B[\frac{k-1}{2}, \frac{n-k+2}{2}]\}^{-1} \Gamma(\frac{n-k+3}{2}) \\ \sum_{\mu=0}^{\infty} \left\{ \frac{2^{2\mu} \Gamma(\mu + \frac{3}{2}) \Gamma(\mu + \frac{k}{2} + \frac{1}{2})}{(2\mu)! \Gamma(\mu + \frac{n}{2} + 2)} \Gamma^{\star}(\mu + \frac{1}{2}, \frac{\Delta}{8}) \right\}.$$

Proof:

$$e(1:2) = \Pr(D_2 \leq 0) = \Pr(D_2^{\star} \leq 0) = \int_{-\infty}^{\frac{\sqrt{\Delta}}{2}} f_{L_{k,n}}(t) dt.$$

The statement of the theorem follows from lemma 4 and the fact that

$$\int_{-\frac{\sqrt{\Delta}}{2}}^{\frac{\sqrt{\Delta}}{2}} t^{2\mu} \exp[-\frac{1}{2}t^2] dt = 2^{\mu-1/2} \Gamma^{\star}(\mu + \frac{1}{2}, \frac{\Delta}{8}).$$

REFERENCES

- ABRAMOWITZ, M. and STEGUN, I.A. (1970). *Handbook of Mathematical Functions*. Dover, New York.
- ANDERSON, T.W. (1966). *An Introduction to Multivariate Statistical Analysis*. Wiley, New York.
- BANDYOPADHYAY, S. (1978). Classification, distributions and admissibility in autoregressive process. *Austral. J. Statist.* **20**, 229-239.
- DAS GUPTA, S. (1973). Theories and methods in classification: a review, *Discriminant Analysis and Applications* edited by T. Cacoullos. Academic Press, New York, pp. 77-137.
- ERDÉLYI, A. et al. (1953). *Higher Transcendental Functions 1*, McGraw-Hill, New York.
- GRADSHTEYN, I.S. and RYZHIK, I.M. (1965). *Tables of Integrals, Series and Products*. Academic Press, New York.
- HUG, G. and STREIT, F. (1976). Linear discriminant analysis when μ_1 and μ_2 are known and Σ is unknown. *Bulletin of the IMS* **5**, 207.
- JOHN, S. (1961). Errors in discrimination. *Ann. Math. Statist.* **32**, 1125-1144.
- KSHIRSAGAR, A.M. (1972). *Multivariate Analysis*. Dekker, New York.
- LACHENBRUCH, P.A. (1975). *Discriminant Analysis*. Hafner, New York.
- STREIT, F. (1977). Identification rules based on partial information on the parameters, *Recent Developments in Statistics* edited by J.R. Barra et al., pp. 797-806.
- STREIT, F. (1979). On the unconditional probabilities of misclassification of an identification rule. *Bulletin of the IMS*, **8**, 113.